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ORIGINAL ARTICLE

# Two-stage heuristic genetic optimization algorithm for multi-UAV logistics task allocation

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**Abstract**
Unmanned Aerial Vehicles (UAVs) have gradually played an increasingly important role in the low-altitude economy. The task allocation problem has become a vital issue in the application of the logistics of groups of low-altitude UAVs, and the focus of UAV logistics enterprises is to improve the system's operation efficiency and economic benefits. Currently, mainstream multi-UAV task allocation algorithms cannot solve the satisfactory task allocation scheme in an acceptable time. The contribution of this paper is the proposal of a new two-stage heuristic algorithm based on a combination of the Hungarian algorithm and an improved genetic algorithm that achieves global performance optimization and obtains an overall superior solution in less time. The experiments showed that the proposed algorithm performed 15.26% better than the traditional genetic algorithm regarding overall task allocation revenue and required 17.6% less calculation time, indicating a better overall solution. The algorithm proposed in this paper provides a new optimization strategy for solving the problem of low-altitude multi-UAV task allocation.

**Keywords** Low-altitude economy · Logistics tasks · Multi-UAV task allocation · Genetic algorithm · Heuristic algorithm

## 1 Introduction

Logistics tasks generally refer to the distribution of goods from the service center to the customer; these tasks include goods distribution, express delivery, drug distribution, and emergency relief material distribution. The primary medium of distribution is vehicles. In 2021, China first proposed the concept of a low-altitude economy in the "National Comprehensive Three-dimensional Transportation Network Planning Outline" [1–3]. To explore new models of low-altitude economic development, the United States [4] and European Union [5] have proposed the Next Generation Air Transportation System (NexGen), Single European Sky Air Traffic Management Research (SESAR), and other projects. Since 2023, the low-altitude economy has become an increasingly strategic emerging industry in China and was initially applied to smart agriculture, smart logistics, tourism, and social governance [6]. As customer demand for the timeliness of logistics has grown higher and higher, the air transportation capacity of unmanned aerial vehicles (UAVs) and their ability to avoid ground transportation have become the focus of attention. For example, in February 2021, China's Meituan UAV research team launched a distribution service for physical users in Shenzhen [7]. In addition, Walmart and DoorDash [8] plan to achieve large-scale UAV logistics scheduling in the United States by 2024. Although the current UAV goods distribution system has progressed, it is limited by a simple point-to-point fixed-route mode. Consequently, the distribution decision-making problem in large-scale complex urban scenarios remains a bottleneck. The distribution of UAV logistics involving multiple tasks still faces issues such as long calculation times, poor solution quality, and weak generalization performance, and related algorithms require improvements [9, 10]. Increasing the efficiency at which UAVs distribute items requires coordinating and linking multiple UAVs. After a user submits an order to a merchant, the distribution system creates a reasonable allocation and schedule according to the current task status and assigns tasks with high distribution pressure or high timeliness requirements for the UAV to complete.

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Therefore, the core advantages of distributing items via UAVs are supplementing the transportation capacity and efficiently allocating tasks [11].

In recent years, scholars have made many achievements in large-scale UAV task allocation. Allocation algorithms can be divided into four categories:

(1) Mathematical programming methods are represented by exhaustive methods, branch and bound algorithms, and dynamic programming methods [12, 13]. For example, Song et al. [14] proposed an approximate two-phase (ATP) method based on the mathematical model of a multi-objective Pareto solution set and successfully applied it to the task allocation problem of small UAV disaster management. In addition, Liu et al. [15] proposed a dynamic programming method for heterogeneous UAVs and constructed a collaborative network based on reinforcement learning to conduct UAV task allocation. This kind of algorithm can play a good optimization effect in small-scale problems, but it cannot obtain a feasible solution in a finite time in large-scale problems.

(2) Heuristic algorithms include the Hungarian, evolutionary, genetic, and tabu search algorithms [16–18]. For example, Rottondi et al. [19] proposed a heuristic-based greedy algorithm to develop a framework for multi-UAV task allocation algorithms designed for disaster response. They verified the effectiveness of their algorithm's capacity to allocate the distribution of relief supplies during floods in San Francisco in 2017. To solve the discrete task allocation problem of the genetic algorithm, Younas et al. [20] proposed a topological crossover operator based on a chaotic sequence, conflict-resolution strategy, and shuffling list. Yu et al. [21] proposed an improved genetic algorithm with novel chromosome coding and unlocking mechanisms for planning cooperative missions of heterogeneous UAVs. Yan et al. [22] used the improved genetic algorithm to solve the problem of multi-UAV task allocation and path planning, and designed an unlocking strategy to imitate the occurrence of infeasible chromosomes. Zhang et al. [23] improved the differential evolution algorithm to solve the path-planning problem of UAVs in complex three-dimensional environments. Combined with the mutation crossover fusion factor, adaptive guidance mechanism, and elite interference mechanism of population classification, the efficiency of the differential evolution algorithm in the UAV path planning problem was improved. Heuristic algorithms usually require more iterations, and the quality of the initial solution has a great influence on the solution effect, which often consumes a lot of computing resources.

(3) Swarm intelligence algorithms include particle swarm optimization (PSO), ant colony optimization (ACO), and whale optimization algorithm (WOA) [9, 24, 25], and they are useful for UAV task allocation algorithms. For example, Wang et al. [26] proposed a multi-objective PSO based on adaptive angle-area partitioning for the multi-UAV task allocation problem. Simulation results demonstrated that the improved algorithm provided a feasible solution for UAVs, exhibiting superior convergence and diversity. Based on many wind field three-dimensional detection models, Li et al. [9] proposed an improved bidirectional discrete honeypot ACO algorithm for the problem of allocating UAV transmission line detection tasks. The swarm intelligence algorithms usually have a fast iteration speed, but its population diversity is relatively poor and it is easy to fall into local optimum.

(4) Negotiation mechanism algorithms include the Contract Net Protocol (CNP) [27, 28]. For example, Zhen et al. [29] proposed an improved cooperative target assignment scheme based on a CNP that considered the total flight distance, flight time, and load rate of a UAV to allocate tasks to the UAV. The experimental results demonstrated that the proposed method achieved a superior distribution balance. Negotiation mechanism algorithms can efficiently assign tasks to UAVs, but it is difficult to optimize the global optimal allocation scheme by focusing on the local task allocation environment.

In general, although mathematical programming methods can accurately solve the optimal allocation scheme, the complexity of the solution increases exponentially with the task scale, and the solution cannot be reached quickly. Although heuristic, swarm intelligence, and negotiation mechanism algorithms can quickly obtain feasible solutions, the quality of the solutions is generally poor and cannot satisfy the actual economic requirements. In addition, most studies use the linear distance between the UAV and the task destination as the cost evaluation standard, but this causes problems such as significant deviations between the expected and actual results and difficulties in applying the standard. Therefore, this paper investigates multi-UAV task allocation algorithms based on two-stage heuristic optimization to achieve a more profitable allocation scheme in a shorter time. This paper primarily addresses the problems from the research described above, and its specific contributions include the following.

- To address the problem of significant deviations between task allocation and actual execution, a method using the actual flight trajectory of UAVs as the distance cost is proposed. The traditional method takes the Euclidean distance as the task cost, ignoring the obstacles in the flight process. In this paper, the total distance of the UAV path planning is used as the task cost evaluation to enhance the feasibility and rationality of the planning results.

- To address the problem of the poor solution quality of traditional algorithms, a genetic optimization method for the conflict resolution mechanism is proposed. In this paper, a dynamic genetic crossover operator using a conflict resolution strategy is designed to more effectively balance the global exploration and local exploitation abilities of the algorithm to satisfy these constraints. In addition, the corresponding optimization methods are designed for the three cases of the number of UAVs and the number of tasks.
- To address the problem of the low computational efficiency of traditional algorithms, a heuristic two-stage optimization method is proposed. Taking advantage of the efficient calculation characteristics of the Hungarian algorithm, the distribution results are used as the initial solution of the improved genetic algorithm, which enhances the initial population quality of the genetic algorithm, improves the optimization performance, and reduces the blind search in the early stage.

The rest of this paper is organized as follows. Section 2 describes the UAV logistics task allocation model, Sect. 3 explains the two-stage genetic algorithm based on the local solution heuristic, Sect. 4 details the experiments conducted and presents their results, and Sect. 5 discusses the conclusions of the paper.

## 2 UAV logistics task allocation model

### 2.1 Problem description

The UAV logistics task refers to the operation of a UAV assigned to a designated place to complete a designated task, which is essential for efficiently scheduling low-altitude transportation of goods [12, 30, 31]. The main advantages of these tasks include high distribution efficiency, low labor costs, and timely delivery when traffic inconveniences cannot be circumvented.

When the UAV performs a task, it needs to travel to the merchant's location from the airport to load the goods, and after the loading is finished, it travels to the customer's location to complete the delivery. During this period, the UAVs must avoid inaccessible environments to ensure flight safety. When the UAV $i$ is assigned to perform task $j$, the actual path length from the airport to the merchant's location is $d_{ij}^1$, the path length from the merchant's location to the customer's location is $d_{ij}^2$, and the path length to return to the airport after unloading the goods is $d_{ij}^3$. The total distance cost $D_{ij}$ of the UAV $i$ performing the task $j$ is expressed by

$$D_{ij} = d_{ij}^1 + d_{ij}^2 + d_{ij}^3 \tag{1}$$

The rapidly exploring random tree (RRT) algorithm [32] offers the advantage of a high calculation speed for a feasible path. Therefore, in this paper, the actual UAV flight path is calculated using the RRT, and the corresponding total path cost $D_{ij}$ is used to evaluate the cost of the UAV $i$ performing the task $j$. Figure 1 shows a schematic diagram of three UAVs performing logistics tasks.

Considering the case of one-time task allocation, $N$ UAVs $U = \{U_1, U_2, \cdots, U_N\}$ and $M$ tasks $T = \{T_1, T_2, \cdots, T_M\}$ are assumed. The goal of task allocation is to, for each task $T_j$ ($j \in [1, M]$), assign only one $U_i$ ($i \in [1, N]$). This can be divided into three cases according to the relative number of tasks and UAVs: (1) $M < N$ (the number of tasks is less than the number of UAVs), (2) $M = N$ (the number of tasks is equal to the number of UAVs), and (3) $M > N$ (the number of tasks is greater than the number of UAVs). The algorithm proposed in this paper is able to construct the corresponding optimization strategies, and the specific method by which this is accomplished is elaborated in Sect. 3.2.1. The goal of solving this problem is to maximize the total revenue of the overall task execution scheme by comprehensively considering the total distance cost, load constraints, and revenue of each UAV for the corresponding task.

### 2.2 Model assumptions and symbol description

The multi-UAV task allocation model constructed in this paper was based on the following assumptions.

- The flight height of the UAV remains within a fixed range and does not exceed the preset height.
- The UAV does not suddenly have an energy shortage during the flight.
- The UAVs fly as particles, and the distance between them and obstacles is controllable.
- During UAV flight, the environmental information is stable and predictable.
- The flight speed of the UAV is known, its motion is uniform, and its position can be obtained in real time.
- The UAV is not subjected to the problem of return charging.
- The UAV is not subjected to the problem of task execution failure caused by spilling goods or losing orders.

The symbol definition of this paper is shown in Table 1.

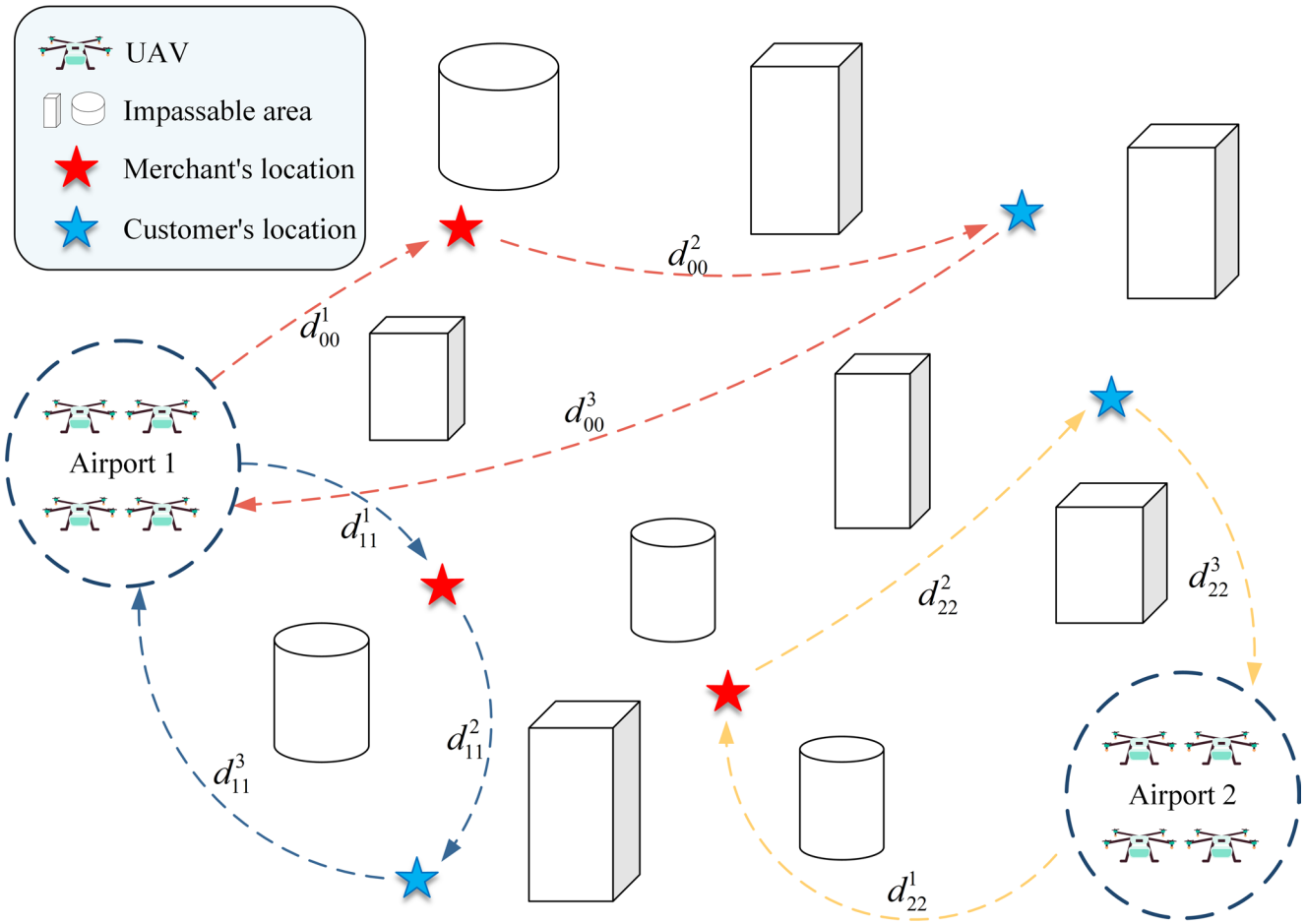



**Fig. 1** Diagram showing the distribution of multi-UAV logistics tasks

**Table 1** Descriptions of variables used in this paper

| Variable | Explanation |
|---|---|
| $N$ | Total number of UAVs. |
| $M$ | Total number of tasks. |
| $U = \{U_1, U_2, \cdots, U_N\}$ | UAV set. |
| $T = \{T_1, T_2, \cdots, T_M\}$ | Task set. |
| $x_{ij} = \{0,1\}$ | $x_{ij} = 1$ means that UAV $i$ performs task $j$, and $x_{ij} = 0$ means that the UAV does not perform the task. |
| $Q_i$ | Maximum flight distance of UAV $i$. |
| $G_i$ | Total load capacity of UAV $i$. |
| $D_{ij}$ | Total path distance of UAV $i$ when performing task $j$. |
| $d_{ij}^1$ | Pick-up distance of UAV $i$ when performing task $j$. |
| $d_{ij}^2$ | Delivery distance of UAV $i$ when performing task $j$. |
| $d_{ij}^3$ | Return distance of UAV $i$ when performing task $j$. |
| $w_j$ | Weight of the cargo for task $j$. |

## 2.3 Constraint functions

The multi-UAV task allocation problem is a combinatorial optimization problem with multiple constraints, which are described as follows.

1. Constraint on the number of tasks performed by a single UAV. A single UAV can perform only one task at a time (i.e., for UAV $i$, the number of tasks performed is either zero or one), as expressed by

$$s.t. \sum_{j=1}^{M} x_{ij} = \{0,1\}, \forall i \in U \tag{2}$$

2. Constraint on the UAV endurance capacity. When a UAV performs its mission, it cannot exceed its total endurance limit, as expressed by.

$$s.t. \sum_{j=1}^{M} x_{ij} \cdot D_{ij} \leq Q_i, \forall i \in U \tag{3}$$

3. Constraint on the UAV load capacity. The weight of the goods involved in the task performed by the UAV cannot exceed its maximum load constraint, as expressed by.

$$s.t. \sum_{j=1}^{M} x_{ij} \cdot w_j \leq G_i, \forall i \in U \tag{4}$$

**Objective function**.

A UAV flies from an airport in a city to a merchant's location. After the cargo is loaded, the UAV travels to the customer's location to unload the cargo, and then it returns to the airport to complete the task delivery. Assuming that UAV $i$ is assigned to perform task $j$, the corresponding decision variable $x_{ij} = 1$, where $x$ represents the task execution of the UAVs. The unit fuel consumption price is denoted as $p_i$, and the reward for completing task $j$ is denoted as $R_j$, the cost of performing this task is $p_i \cdot \left(d_{ij}^1 + d_{ij}^2 + d_{ij}^3\right)$, and the net benefit of completing task $j$ is $R_j - p_i \cdot \left(d_{ij}^1 + d_{ij}^2 + d_{ij}^3\right)$. Therefore, the total revenue $P$ for $N$ UAVs to complete $M$ tasks can be expressed as

$$\begin{aligned} P &= \max_x \left[\sum_{i=1}^{N} \sum_{j=1}^{M} x_{ij} \cdot \left(R_j - p_i \cdot \left(d_{ij}^1 + d_{ij}^2 + d_{ij}^3\right)\right)\right] \\ s.t. &\sum_{j=1}^{M} x_{ij} = \{0,1\}, \forall i \in U \\ &\sum_{j=1}^{M} x_{ij} \cdot D_{ij} \leq Q_i, \forall i \in U \\ &\sum_{j=1}^{M} x_{ij} \cdot w_j \leq G_i, \forall i \in U \end{aligned} \tag{5}$$

The goal of algorithm optimization is to find a combined allocation scheme of UAVs and tasks to maximize the total revenue $P$ under the premise of satisfying the constraints of UAVs.

## 2.4 Model analysis and solving algorithms

Multi-UAV task allocation is a single-objective multi-constraint optimization problem that maximizes the objective function (i.e., $P$ in Eq. (5)) while satisfying the UAV constraints. Because of its multiple constraints, strong coupling, and nonlinear characteristics [33, 34], the difficulty of solving the problem increases exponentially as the number of tasks increases, and it cannot be solved in polynomial time. Thus, it is an NP-hard problem [35]. The mainstream allocation algorithms and their advantages and disadvantages are summarized in Table 2.

The Hungarian algorithm can produce a good solution when the task scale is small, but when the task scale increases or the gap between the number of UAVs and the number of tasks is large, the quality of the solution is poor. Although the heuristic algorithm has a good solution effect, it requires more iterations, is sensitive to the initial solution, and has poor early optimization ability. The contract

**Table 2** The advantages and disadvantages of the mainstream allocation algorithms

| Algorithm typology | Authors | Time | Utilized technique | Advantages | Disadvantages |
|---|---|---|---|---|---|
| Optimization algorithm | Li et al. [36] | 2023 | Numerical optimization. | The algorithm can converge to the optimal solution. | The scale of the problem is small, and it is difficult to solve large-scale problems. |
| | Dong et al. [37] | 2024 | Improved Hungarian algorithm, multi-view clustering. | They have faster solving speed. | The quality of the solution is poor. |
| | Samiei et al. [38] | 2024 | Improved Hungarian algorithm. | | |
| | Zhang et al. [39] | 2023 | Improved Hungarian algorithm, distributed control. | | |
| | ChiJen et al. [40] | 2023 | Weighted Hungarian algorithm. | The algorithms can solve the problem of large-scale task allocation. | The algorithms have high computational complexity. |
| | Zhang et al. [41] | 2024 | Improved Hungarian algorithm. | | |
| Heuristic algorithm | Qin et al. [42] | 2024 | Improved differential evolutionary algorithm. | The algorithms have good solution performance. | The algorithms require a large number of iterations. |
| | Wang et al. [26] | 2023 | Improved particle swarm optimization. | | |
| | Yu et al. [21] | 2022 | Improved genetic algorithms. | | |
| Contract network protocol | Zhen et al. [29] | 2021 | Improved contract network protocol. | The algorithms have faster solution speed. | The solution quality of the algorithms is poor. |
| | Yan et al. [27] | 2021 | | | |
| The algorithm in this paper | | | Two-stage heuristic optimization. | The algorithm can find a better solution in a shorter time. | The algorithm solves better on large scale problems and has average solving ability on small scale. |

net protocol has a fast solution speed, but it is generally unable to find the optimal solution. Therefore, based on the characteristics of the UAV task allocation problem and the advantages of each algorithm, this paper designed a two-stage genetic algorithm based on a local solution heuristic combined with the Hungarian and genetic algorithms to solve the multi-UAV task allocation problem. The algorithm can improve the solution speed while ensuring the solution quality.

## 3 Two-stage genetic algorithm based on local solution heuristic

The algorithm proposed in this paper is divided into two stages. The first stage is the heuristic process based on the allocation results obtained by the Hungarian algorithm (HA) for a local search. The second stage is the global optimization process, in which the improved genetic algorithm is used to further iteratively optimize the local solution of the HA. The two-stage genetic algorithm based on local solution heuristic (TGALS) not only offers a better solution quality but also a faster solution speed, which effectively improves the feasibility, robustness, and efficiency of multi-UAV task allocation.

### 3.1 Stage 1: Hungarian algorithm for heuristic process

The HA [43] is a commonly used combinatorial optimization algorithm based on the sufficiency proof idea of the Hall theorem [44], and it is designed to find the minimum matching problem of the cost matrix utilizing an augmented path. The goal of applying HA to multi-UAV task allocation is transforming the task into the assignment of UAVs. The following four steps were designed based on the concept of a trial assignment.

Step 1. Assuming $N$ UAVs and $M$ tasks, an $N \times M$ cost matrix $H$ is constructed based on the actual flight cost of each UAV performing each task, where each row represents the UAV number and each column represents the task number. If $N \neq M$, the corresponding rows or columns are filled with 0 elements, enabling the cost matrix $H$ to remain square. Assuming $N \geq M$, the constructed cost matrix is expressed as

$$H = \begin{bmatrix} S_{11} & \cdots & S_{1N} \\ \vdots & \ddots & \vdots \\ S_{N1} & \cdots & S_{NN} \end{bmatrix} \tag{6}$$

.

Step 2: Reduce the cost matrix $H$, subtract the minimum value of each row from the corresponding rows, and subtract the minimum value of each column from the corresponding columns.

Step 3: Perform the first assignment. The assignment principle covers all zero elements in the cost matrix $H$ with as few lines as possible. If the number of lines used is equal to $N$, the algorithm stops and outputs the assignment result. If it is less than $N$, Step 4 is performed.

Step 4: If the number of lines used is less than $N$, calculate the minimum value of the uncovered elements, subtract the minimum value from all elements not covered by the line, add the minimum value to the repeatedly covered elements, update the cost matrix $N$, and return to Step 3.

Figure 2 shows a schematic diagram of the UAV task allocation problem in which the HA is to solve four tasks. The task allocation scheme was as follows: $U_1$ performs $T_4$, $U_2$ performs $T_2$, $U_3$ performs $T_3$, $U_4$ performs $T_1$.

### 3.2 Stage 2: Multi-UAV task assignment genetic algorithm based on conflict resolution mechanism

The genetic algorithm (GA) is an intelligent optimization algorithm designed to simulate the genetic evolution of biological populations. Population evolution is realized through a crossover operator, mutation operator, and selection operation. It has the advantages of a strong global search ability, excellent versatility, and easy implementation [45, 46]. However, the traditional GA has certain limitations. When dealing with constraint problems, a large search space leads to problems such as low efficiency in searching for feasible solutions, small chromosome differences in the later stages of the algorithm, and premature convergence [47]. To address the problem of multi-UAV task allocation, it is necessary to improve and optimize the coding method and crossover operator of the GA to adapt to the domain of this paper because the number of UAVs and tasks are typically different and the scale is large. Accordingly, this paper designed an optimization concept using a conflict resolution mechanism, which is described as follows.

#### 3.2.1 Coding design and conflict resolution mechanism

In the UAV task allocation problem, it is assumed that the population contains $ps$ chromosomes, representing $ps$ feasible schemes for multi-UAV task allocation [48]. Each chromosome is composed of genes with a fixed length, and the number of genes is the same as the number of tasks to be performed. The value of each gene is an integer between $[1, N]$, indicating that the task is to be performed by the UAV with the corresponding number label. Three main

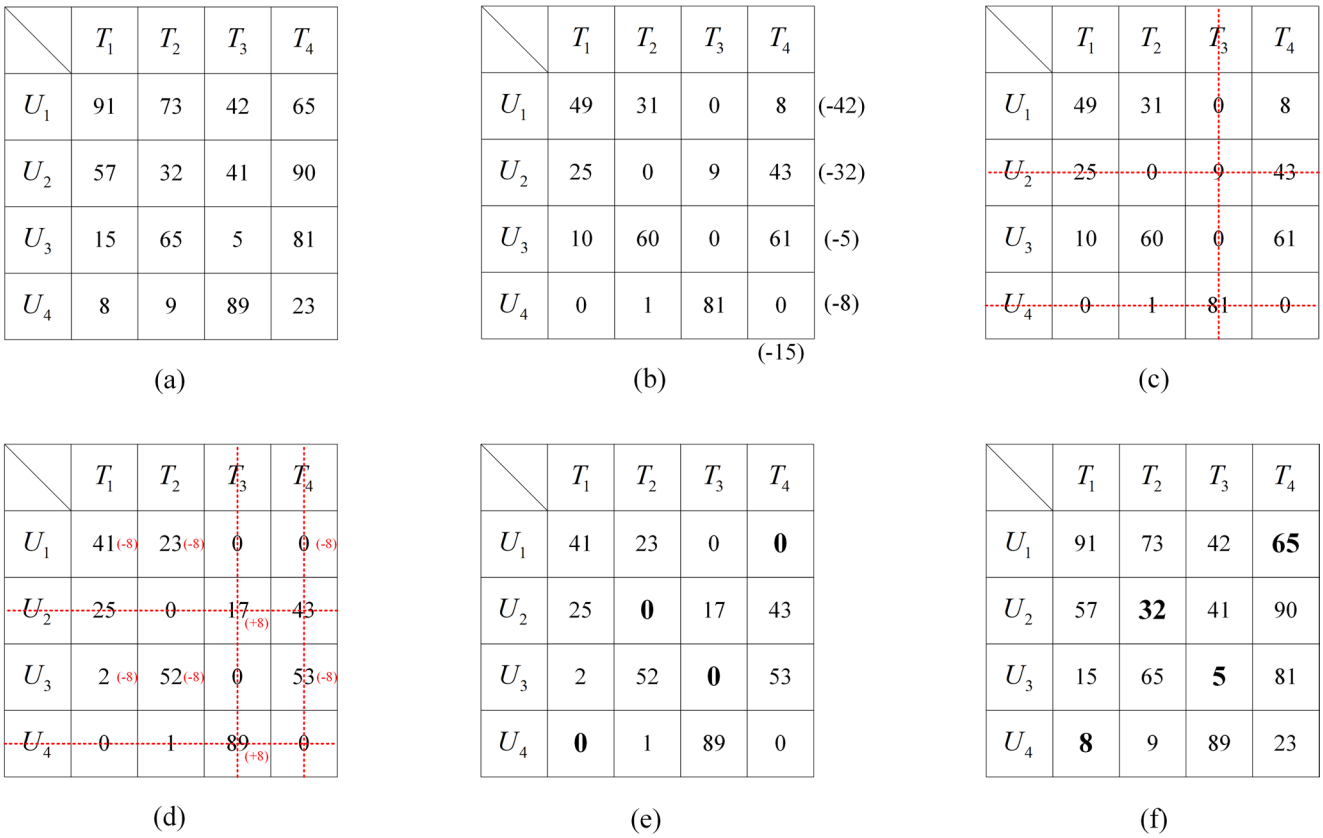



**Fig. 2** Diagram showing the multi-UAV task allocation process conducted using the HA

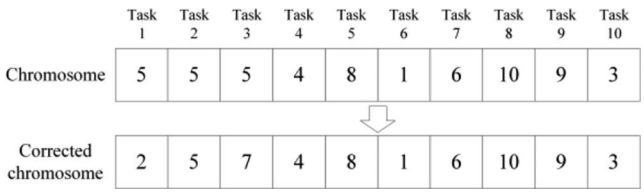



**Fig. 3** Chromosome coding when $N = M$

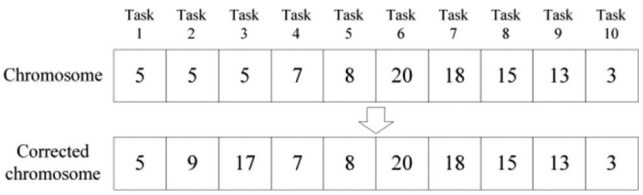



**Fig. 4** Chromosome coding when $N > M$

situations are possible for the coding design of multi-UAV task allocation, and they are described as follows.

1. $N = M$: the number of UAVs is the same as the number of tasks, and all UAVs are sent to perform the tasks. The conflict resolution mechanism corrects a chromosome if the same UAV performs multiple tasks. The coding method for a single chromosome, assuming 10 tasks and 10 UAVs, is shown in Fig. 3. The figure indicates that UAV 5 is assigned to perform tasks 1, 2, and 3 simultaneously, and a task will be randomly selected to be continued by UAV 5. The idle UAVs 2 and 5 perform the remaining tasks sequentially to ensure the feasibility of the chromosome.

2. $N > M$: the number of UAVs exceeds the number of tasks. In this case, only a few UAVs perform the tasks. If a UAV performs multiple tasks, the conflict resolution mechanism is expressed as follows: a task is randomly retained to continue to be performed by this UAV, and another UAV is randomly selected from the idle UAVs to perform the remaining tasks. Assuming 20 UAVs and 10 tasks, the coding method is illustrated in Fig. 4.

3. $N < M$: the number of UAVs is less than the number of tasks. All UAVs are sent to perform the tasks, and some tasks cannot be performed. The unexecuted tasks correspond to the chromosome gene coding set to -1. If a situation in which a UAV performs multiple tasks occurs, the conflict resolution mechanism is expressed as follows: one task is randomly retained to continue to be performed by the UAV, and the remaining task genes are set to -1. Assuming 10 tasks and five UAVs, the coding method is shown in Fig. 5.

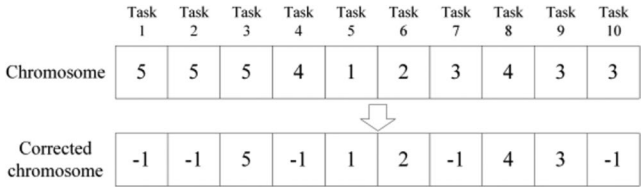



**Fig. 5** Chromosome coding when $N < M$

In the initialization phase of the algorithm, $ps$ chromosomes are randomly generated to form the initial population. The HA solution is added to the current population and serves as an elite individual that guides the evolution of other individuals in the population. The fitness evaluation function calculates the fitness value of each chromosome in the population.

### 3.2.2 Fitness evaluation function

The chromosome corresponds to a multi-UAV task allocation scheme. In this paper, the fitness value of a single chromosome is defined as the total income value of the task allocation scheme. Using Fig. 5 as an example, the chromosome can be expressed as UAV 5 assigned to perform task 3, and UAV 1 assigned to perform task 5. Equation (5) was used to calculate the total income $P$ of the allocation scheme, which is the fitness value of the chromosome. The greater the fitness value, the greater the income of the allocation scheme and the value of the chromosome.

### 3.2.3 Improved crossover operator

In the crossover operation of the traditional GA, the feasibility of offspring individuals is not considered. Adding infeasible solutions to the evolution of the offspring will cause the population as a whole to slowly evolve, and it is difficult to efficiently find a better solution. As shown in Fig. 6, assuming $N = M$, UAV 5 in the offspring performs the first and third tasks simultaneously. If the offspring are not processed and continue to participate in subsequent iterations, the infeasibility of the entire population will increase, and searching for the best feasible solution will be difficult. Therefore, this paper designed a conflict resolution strategy based on dynamic adjustment to correct the offspring that do not satisfy the constraints in the iterative process.

In the early stages of the algorithm optimization process, it is necessary to quickly locate the potential region in the search space. In the later stages, it is necessary to perform fine development near the local region searched to obtain a better solution. In this paper, a dynamic adjustment strategy was designed to select the objects to be used in conflict resolution. The adjustment equation is expressed as

$$TA = \begin{cases} B_g, & if\left(r_1 > \frac{t}{t_{max}}\right) \\ I_{uav}, & otherwise \end{cases} \tag{7}$$

where $TA$ represents the selected auxiliary object, $B_g$ represents the elite individual in the current population, $r_1$ represents a random number between $[0,1]$, $t$ represents the current number of iterations, $t_{max}$ represents the maximum

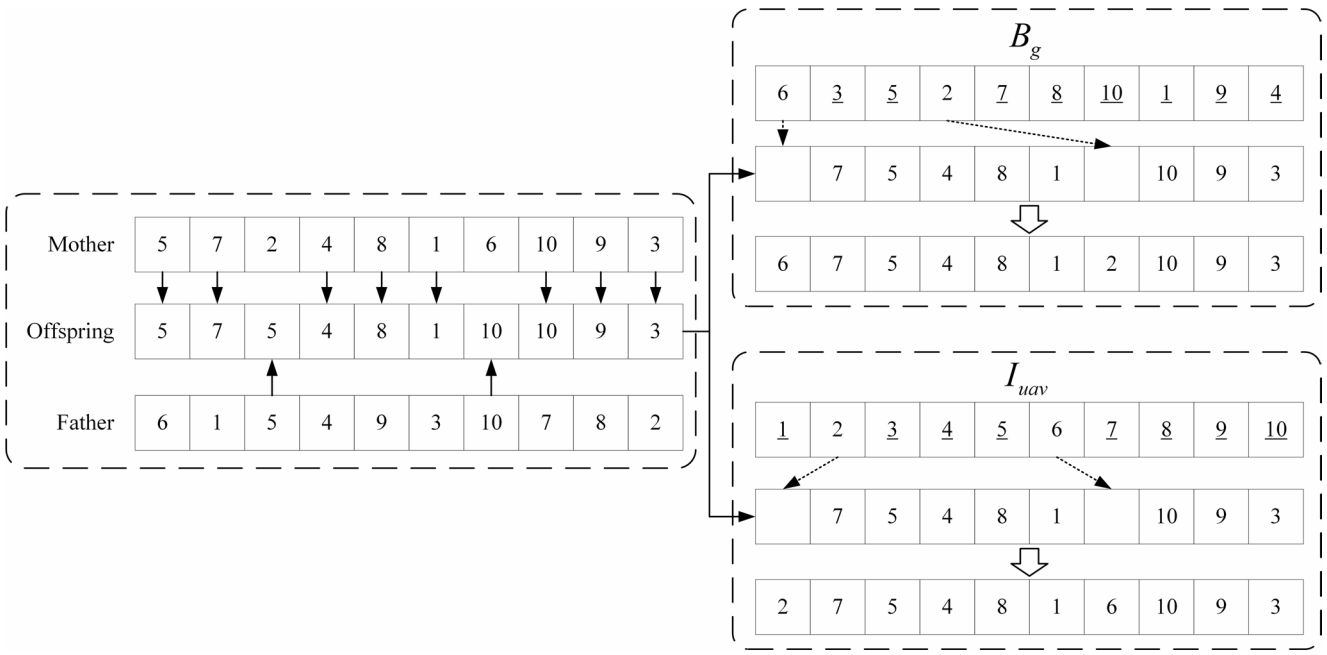



**Fig. 6** Flow chart of the improved cross-operator

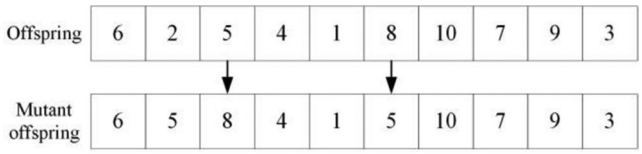



**Fig. 7** Mutation operator

number of iterations, and $I_{uav}$ represents a list of UAVs that have not been dispatched to perform tasks. By utilizing Eq. (7), the object used in the conflict resolution can be dynamically adjusted iteratively to carry out the correction operation on the infeasible offspring. According to Eq. (7), in the early stage of the algorithm's search, selecting elite individuals for correct to improve the speed of finding potential regions is highly probable; in the later stage, there is a high probability of using idle UAVs to correct, enhance the diversity of the generated offspring population, reduce the possibility of falling into a local optimum, and effectively balance the global exploration and local exploitation abilities of the algorithm.

#### 3.2.4 Mutation operator

In the process of biological evolution in nature, individuals have a certain probability of mutation, which is expressed as the change of chromosome corresponding gene loci in GA to enhance the diversity of the population. The text adopts a random mutation strategy, assuming $N = M$, and the mutation process is shown in Fig. 7.

### 3.3 Algorithm flow

TGALS uses the local solution of the HA as an elite individual of the improved genetic algorithm to guide the population evolution, which improves the computational efficiency and enhances the optimization performance of TGALS. The main framework of TGALS is shown in Fig. 8.

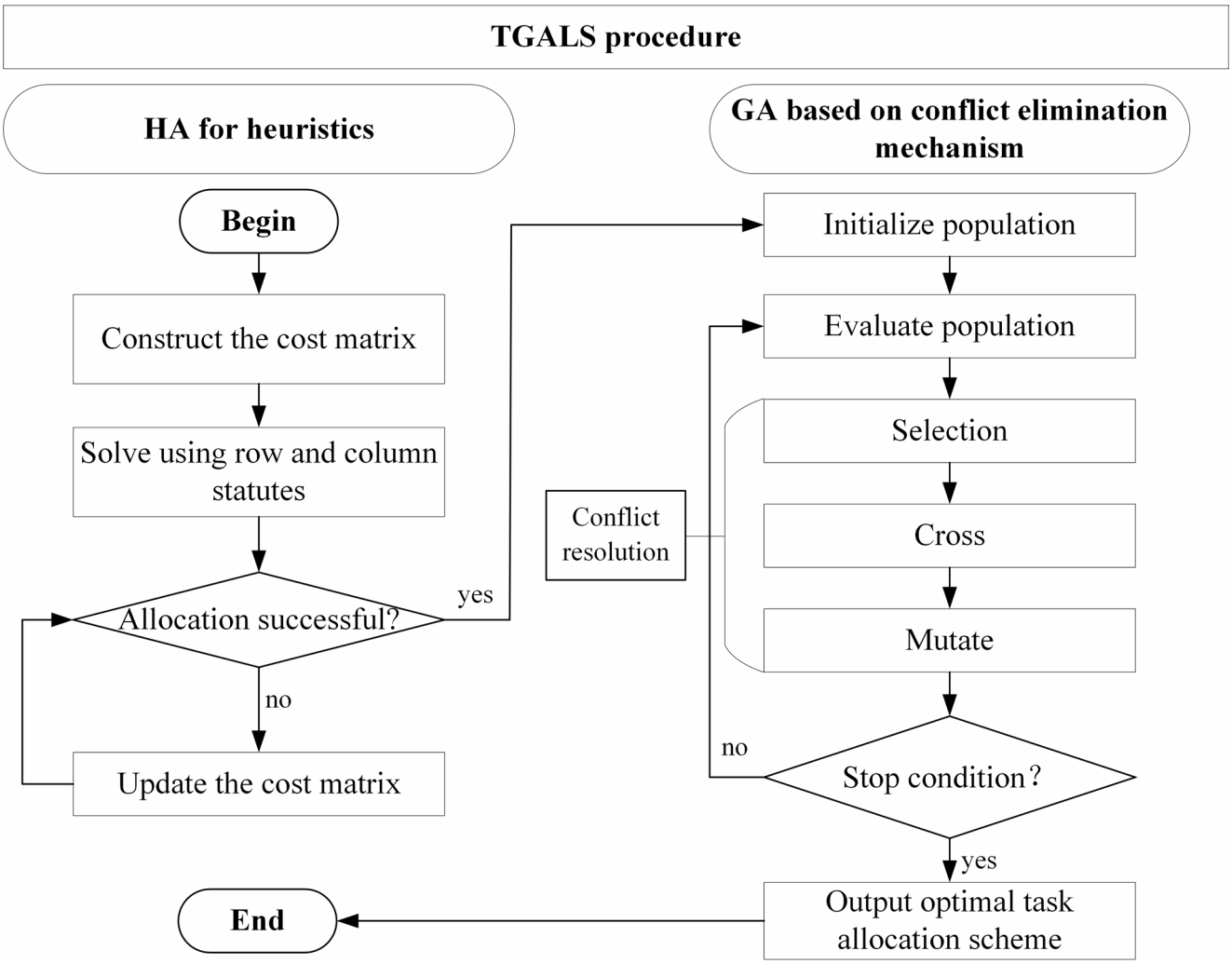



**Fig. 8** TGALS flow chart

The algorithm first uses the HA to pre-allocate the overall task, and the obtained allocation scheme is introduced into the GA as an elite individual to guide the population evolution. A GA with a conflict resolution mechanism is used to further develop the search in the current local area. Finally, the global task allocation scheme with the highest total revenue is sought by setting the maximum number of iterations and using crossover, mutation, and selection operations. The pseudocode for this algorithm is presented in Algorithm 1.

**Algorithm 1** TGALS

**Input:** $T, U, t_{max}$

**Output:** Task allocation results

$LocalGbest \leftarrow HA(T,U)$;
$t \leftarrow 0$;
$P(t) \leftarrow InitializePopulation(LocalGbest)$;
$Evaluate(P(t))$;
$B_g \leftarrow Select(P(t))$;
$while(t < t_{max})$
  $P(t) \leftarrow Selection(P(t))$;
  $P(t) \leftarrow Crossover(P(t))$;
  $P(t) \leftarrow Mutation(P(t))$;
  $Evaluate(P(t))$;
  $for\ each\ p \in P(t)\ do$
    $if(p > B_g)$
      $B_g = p$;
  $end\ for$
  $t++$;
$end\ while$

The traditional multi-UAV task allocation algorithm considers the Euclidean distance between the starting point and the endpoint as the distance cost of the UAV, which causes the actual cost to be significantly different from the expected cost. The RRT [32] can effectively and quickly explore a feasible path in space by randomly expanding the tree, and it performs well in a high-dimensional space. Therefore, to improve the feasibility and rationality of the allocation algorithm, this paper used the RRT algorithm to calculate the actual flight cost of a UAV, which is used in the fitness evaluation process of a chromosome.

For example, as shown in Fig. 9, there are 5 UAVs and 10 tasks, each of which has different benefits, a situation where $N < M$. The first stage of TGALS uses HA to assign tasks, corresponding to Sect. 3.1. At this time, the total income of the allocation scheme is 2389.36 yuan. In the second stage of TGALS, based on the first stage, the improved GA is used to optimize the allocation scheme further, and a total income of 2812.63 yuan is obtained, an overall increase of 17.7%. The final allocation scheme is $U_1$ performs $T_{10}$, $U_2$ performs $T_3$, $U_3$ performs $T_4$, $U_4$ performs $T_7$ and $U_5$ performs $T_6$.

# 4 Experiment, results, and analysis

## 4.1 Experimental design

Given the feasibility, robustness, and efficiency of the TGALS algorithm for actual task allocation, this paper designed the following four groups of experiments involving an actual urban logistics task allocation scenario with 100 UAVs to verify the performance of TGALS.

1. Feasibility experiment. When the number of tasks was 2, 5, 10, 20, 50, or 100, the feasibility of the task allocation scheme calculated by TGALS and the flight conditions of the UAV in the actual execution of the task were tested and analyzed.
2. Robustness experiment. When the number of tasks was 20, 40, 60, 80, 100, 120, 140, 160, 180, and 200, TGALS was compared to the mainstream task allocation algorithms to verify its generalization performance and the increased total income of its allocation scheme relative to those of the mainstream algorithms.
3. Efficiency experiment. When the number of tasks was 100, 200, 300, 400, 500, 600, 700, 800, 900, and 1000, the average running times of TGALS and the mainstream algorithms were measured to verify the computational efficiency of TGALS compared to the GA after using the two-stage optimization strategy.
4. Ablation experiment. Through ablation experiments, the effectiveness of the two-stage strategy and conflict resolution strategy proposed in this paper were verified.

## 4.2 Experimental data and environment

The experimental area was selected from urban areas of Zhengzhou City, Henan Province, China. The longitude range was 113.6626° − 113.6819° and the latitude range was 34.7768° − 34.7982°. The actual area was approximately 2.441 km×2.121 km, and the height of the experimental area was 60–220 m. The original environmental data included a three-dimensional model constructed using oblique photogrammetry, which is shown in Fig. 10(a). Urban scene data were modeled using a grid [49], and a spatial voxel model was established. The voxels were used to identify areas inaccessible to UAVs. The gridded voxel model is shown in Fig. 10(b).

Five UAV airports were used in the experimental region. The airports were designated as Airports 1–5, as shown in Fig. 10(c), and their latitude and longitude coordinates were (113.6751°E, 34.7778°N), (113.6739°E, 34.7788°N), (113.6657°E, 34.7883°N), (113.6698°E, 34.7946°N), and (113.6799°E, 34.7950°N), respectively. Each airport had 20 UAVs, for a total of 100. According to the flight data of

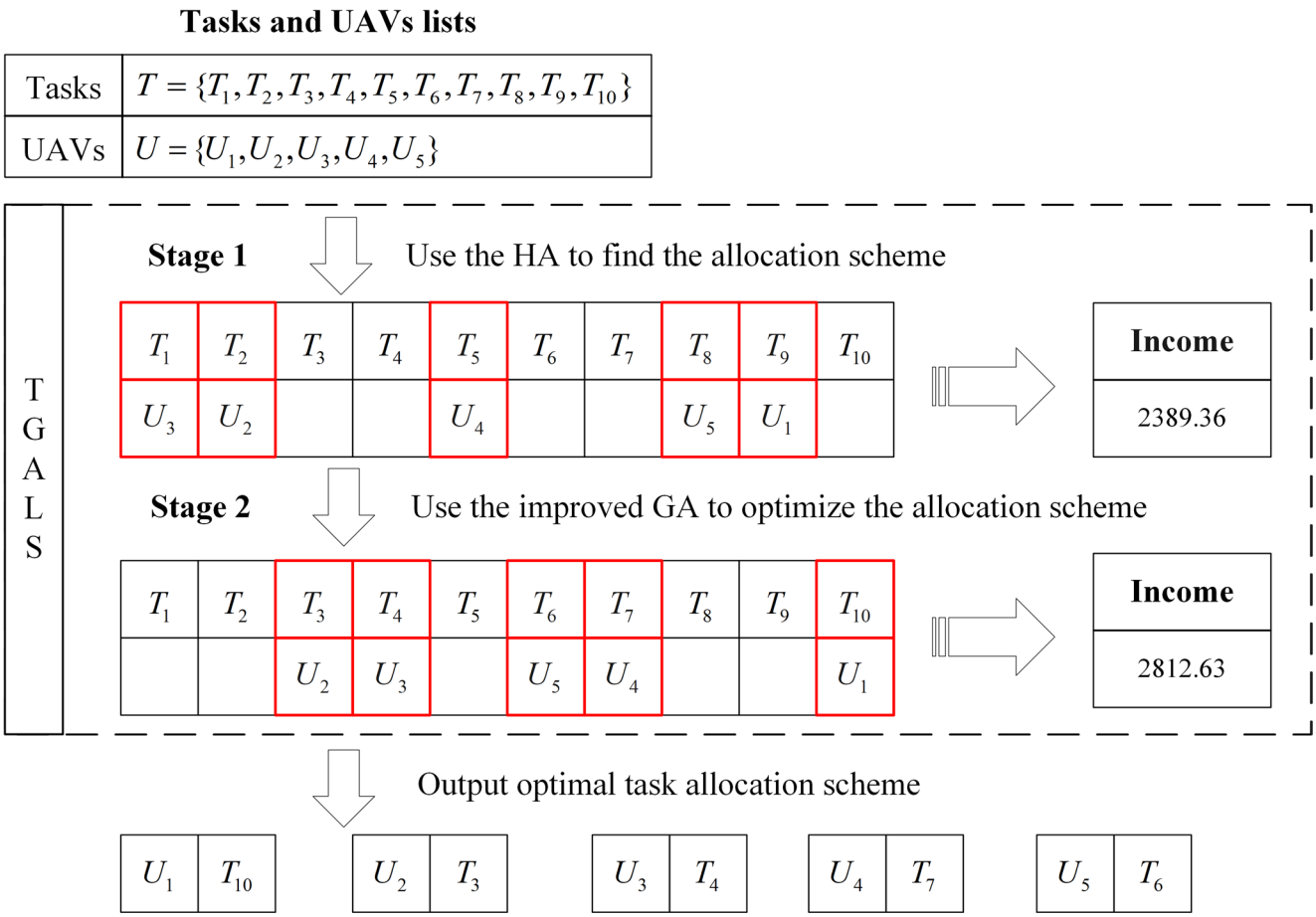
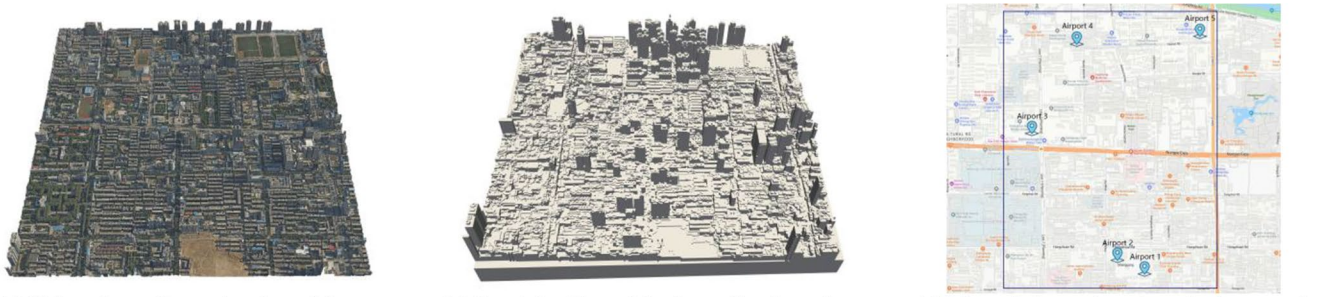



**Fig. 9** A simple example of TGALS task allocation is illustrated when $N < M$



(a) Urban three-dimensional model constructed using oblique photogrammetry

(b) Spatial grid model of voxelization of urban elements

(c) Map of airport locations in the experimental region

**Fig. 10** Data on urban regions in Zhengzhou City, Henan Province, China

$U_i$, the maximum load weight of $U_i$ was $G_i = 14.5$ kg, the maximum range of a full load was $Q_i = 9$ km, and the standard cruise speed was 10 m/s. In the experiment, logistics task $T_i$ was randomly generated in the experimental area, and the merchant's and customer's locations did not lie outside the experimental area.

The processor used to run the experiment was the 11th Gen Intel (R) Core (TM) i5-1135G7 @ 2.40 GHz, with a running memory of 16 GB. To implement the algorithm, the C++programming language was used in the compilation environment of Visual Studio 2022 on the Windows 10×64 operating system.

### 4.3 Experimental steps

(1) Generation of experimental data and element structures. The element attribute structure of a single logistics task is $T_i = \langle id_i, p1_i, p2_i, w_i, R_i, allocated_i \rangle$, where $id_i$ is the

task number, $p1_i$ and $p2_i$ are the business and customer locations ( $p1_i$ and $p2_i$ do not lie outside the experimental area), respectively, $w_i$ is the weight of the goods, $R_i$ is the amount of income earned by completing the task ( $R_i \in [50, 500]$), $allocated_i$ indicates whether the task is allocated ( $allocated_i = \{true, false\}$), the weight of the goods $w_i \in [0.5, 10]$ and its unit is the kilogram (kg), and the unit of currency is RMB (yuan). The element attribute structure of a single UAV is $U_i = \langle id_i, p_i, airport_i, T_i, isworking_i \rangle$, where $id_i$ denotes the UAV number, $p_i$ denotes the UAV location, $airport_i$ denotes the UAV airport, $T_i$ denotes the task element assigned to the UAV, and $isworking_i$ denotes the working state of the UAV ( $isworking_i = \{true, false\}$).

(2) Steps taken in the feasibility experiment. TGALS was used to solve the task allocation scheme, and the actual task execution of the UAV was analyzed. Equation (1) was used to calculate the total distance cost, and Eq. (5) was used to calculate the total income of the task allocation scheme. After the calculations were completed, the actual execution of the UAV flight when performing the task was analyzed.

(3) Steps taken in the robustness experiment. The performances of the HA [43], OSDAA [41], VsC_mH [37], DMCHBA [38], CNP [29], PSO [26], IDE [42], DE_best_2 [50], and GA [20] were compared to that of the TGALS, the parameter settings of the algorithms were shown in Table 3. Where $ps$ is the population size, $maxGen$ is the maximum number of iterations, $c_1$ is the individual learning factor, $c_2$ is the global learning factor, $w$ is the inertia weight interval, $mu$ and $cr$ are the mutation probability and crossover probability, respectively. The control variable method was used in the experiment. Each group of experiments was run 25 times independently, and the total income generated by each algorithm was calculated. Wilcoxon rank-sum test was used to verify the differences in the final allocation results [51], and the task allocation results of the various algorithms were comprehensively analyzed.

(4) Steps taken in the efficiency experiment. The average running times of the five algorithms (i.e., PSO, IDE, DE_best_2, GA, and TGALS) were calculated when the number of large-scale tasks was large. The computational efficiency of TGALS was comprehensively analyzed, and the difference between the computational times of TGALS and GA were compared to verify the effectiveness of the two-stage optimization strategy.

**Table 3** Parameter setting of comparison algorithms

| Algorithm | Experimental parameters |
|---|---|
| PSO | *ps=100; maxGen=100; $c_1$=2.5; $c_2$=4; w=[0.2,0.8]* |
| IDE | *ps=100; maxGen=100; cr=0.8; mu=0.2* |
| DE_best_2 | *ps=100; maxGen=100; cr=0.8; mu=0.2* |
| GA | *ps=100; maxGen=100; cr=0.8; mu=0.2* |
| TGALS | *ps=100; maxGen=100; cr=0.8; mu=0.2* |

(5) Steps taken in the ablation experiment. By using the control variable method, the TGALAS without two-stage optimization (TGALS_A) and the TGALS without conflict resolution (TGALS_B) were compared with TGALS to verify the effectiveness of the proposed two strategies.

## 4.4 Experimental results and analysis

### 4.4.1 Feasibility experiment

Table 4 shows the mission assignment and execution of TGALS when the number of missions was 2, 5, 10, 20, 50, and 100. The merchant and customer locations were the pick-up and harvest points, respectively, and local coordinates were used in the experiment.

Figure 11 shows the route map of the UAV flight when it performed an actual task. Figure 11(a), (b), (c), (d), (e), and (f) show the actual flight trajectories of the UAV when the number of tasks was 2, 5, 10, 20, 50, and 100, respectively. Lines of the same color represent the flight trajectories of a single UAV.

The results shown in Table 4; Fig. 11 indicate that the algorithm can formulate a reasonable task allocation scheme for UAVs with different task numbers. Figure 11 indicates that the UAV took off from the airport up to a fixed height and traveled to the merchant's location to pick up the goods. After loading the goods, it flew to the customer's location to unload the goods. Once this task was completed, it returned to the airport from which it took off. In addition, the UAV avoided impassable areas to ensure flight safety during the mission. Path planning also solved the problem of collision between UAVs [52]. However, the main focus of this paper was the task allocation of UAVs.

### 4.4.2 Parameter selection of TGALS

In this paper, the cross-validation method was used to determine the size of the mutation probability $mu$ and the crossover probability $cr$. The probability interval was [0,1], as shown in Fig. 12. From the statistical results, it can be concluded that when $mu = 0.2$, $cr = 0.8$, the solution of the algorithm is optimal. The maximum number of iterations also has a great influence on the solving ability of the algorithm. Although a large number of iterations can bring good solving ability, it will also consume more computing resources. In this paper, the number of iterations was analyzed, and the inflection point of solving quality growth was selected as the maximum number of iterations of TGALS, that is 100 times.

**Table 4** Implementation of the task allocation scheme in the feasibility experiment

| Task number | UAV | Airport | Task | Merchant's location | | Customer's location | | Income (yuan) | Distance (m) |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 1 | 1 | 1 | 202.4 | 29.6785 | 29.792 | 49.367 | 448.247 | 352.073 |
| | 4 | 4 | 0 | 195.846 | 158.435 | 190.43 | 150.29 | 541.582 | 171.867 |
| 5 | 1 | 1 | 0 | 235.588 | 61.8481 | 36.448 | 57.4958 | 519.868 | 413.582 |
| | 2 | 2 | 3 | 90.1521 | 141.742 | 98.404 | 143.875 | 468.281 | 126.945 |
| | 3 | 3 | 1 | 62.8619 | 203.142 | 166.77 | 59.6583 | 457.531 | 375.59 |
| | 6 | 1 | 4 | 124.968 | 13.831 | 132.503 | 115.065 | 436.331 | 204.747 |
| | 8 | 3 | 2 | 44.7523 | 236.925 | 96.410 | 198.692 | 496.633 | 130.185 |
| 10 | 0 | 0 | 5 | 201.214 | 31.3778 | 158.795 | 119.597 | 414.544 | 263.377 |
| | 1 | 1 | 0 | 169.032 | 34.8315 | 114.856 | 160.35 | 532.557 | 320.935 |
| | … | … | … | … | … | … | … | … | … |
| | 12 | 2 | 6 | 23.552 | 210.39 | 51.3139 | 182.051 | 510.091 | 120.502 |
| | 17 | 2 | 9 | 88.9777 | 59.0677 | 84.838 | 139.512 | 581.898 | 242.468 |
| 20 | 0 | 0 | 17 | 207.17 | 115.418 | 193.187 | 85.8565 | 497.834 | 240.646 |
| | 1 | 1 | 0 | 218.195 | 163.155 | 118.475 | 43.1655 | 425.497 | 350.177 |
| | … | … | … | … | … | … | … | … | … |
| | 28 | 3 | 18 | 14.9971 | 189.967 | 187.406 | 219.907 | 457.761 | 350.716 |
| | 29 | 4 | 16 | 239.548 | 176.155 | 131.706 | 121.463 | 484.11 | 318.391 |
| 50 | 0 | 0 | 18 | 175.049 | 53.9739 | 243.064 | 156.677 | 544.432 | 348.005 |
| | 1 | 1 | 0 | 112.087 | 82.676 | 28.0129 | 35.7508 | 409.349 | 265.122 |
| | … | … | … | … | … | … | … | … | … |
| | 73 | 3 | 40 | 100.331 | 171.014 | 83.496 | 157.649 | 562.266 | 115.206 |
| | 78 | 3 | 45 | 22.9763 | 201.057 | 171.358 | 135.132 | 550.134 | 341.263 |
| 100 | 0 | 0 | 47 | 175.448 | 13.3435 | 201.774 | 44.3022 | 483.795 | 129.959 |
| | 1 | 1 | 1 | 52.5966 | 34.1873 | 219.733 | 181.367 | 408.19 | 486.807 |
| | … | … | … | … | … | … | … | … | … |
| | 98 | 3 | 77 | 112.469 | 146.461 | 139.224 | 98.7927 | 562.217 | 243.898 |
| | 99 | 4 | 85 | 149.302 | 135.148 | 224.626 | 146.87 | 582.937 | 278.028 |

#### 4.4.3 Robustness experiment

Table 5 lists the total average income and variance of the HA, OSDAA, VsC_mH, DMCHBA, CNP, PSO, IDE, DE_best_2, GA, and TGALS algorithms. The results of the Wilcoxon rank-sum are marked as coarsening. The confidence level of the hypothesis test used was $\alpha = 0.05$, indicating the decision risk of the hypothesis test. The (+/-/=) in the comparison results indicate that TGALS is significantly better than, worse than, and indistinguishable from the compared algorithms.

To more intuitively analyze the various results shown in Table 5; Fig. 13 shows a comparison of the allocation results of the ten algorithms. The bars indicate the total incomes generated by the ten algorithms (corresponding to the left y-axis). The dashed lines show the percentage increase in the income achieved by TGALS compared to the other algorithms (corresponding to the right y-axis). Overall, the total revenue generated by TGALS was significantly higher than those of HA, OSDAA, VsC_mH, DMCHBA, CNP, PSO, IDE, DE_best_2, and GA. The increase was more obvious as the number of tasks increased, with average increases of 8.67%, 1.97%, 1.97%, 2.5%, 2.85%, 13.59%, 10.85%, 10.91%, and 6.54% respectively. When the number of tasks exceeded 100, all the UAVs were sent to perform the tasks. The nature of the problem changed from tasks selecting UAVs to UAVs selecting tasks, and the complexity of the solution increased significantly. As shown in Fig. 13, when the number of tasks was large, TGALS solved the task allocation scheme with a higher total revenue than the other algorithms.

Figure 14 shows box plots of the 25 experimental allocation results for the ten algorithms when the number of tasks was 20, 100, 120, and 200, corresponding to panels (a), (b), (c), and (d) in Fig. 14, respectively. In the box plots, the horizontal lines represent, from top to bottom, the maximum, upper quartile, median, lower quartile, and minimum values. The top and bottom edges of the boxes mark the upper and lower quartiles, respectively. The taller the box, the more dispersed the data distribution, and the lower the box, the more concentrated the data distribution. The box plots indicate that the income generated by TGALS was significantly higher than those of HA, OSDAA, VsC_mH, DMCHBA, CNP, PSO, IDE, DE_best_2, and GA and that the fluctuations in the 25 experimental results were small.

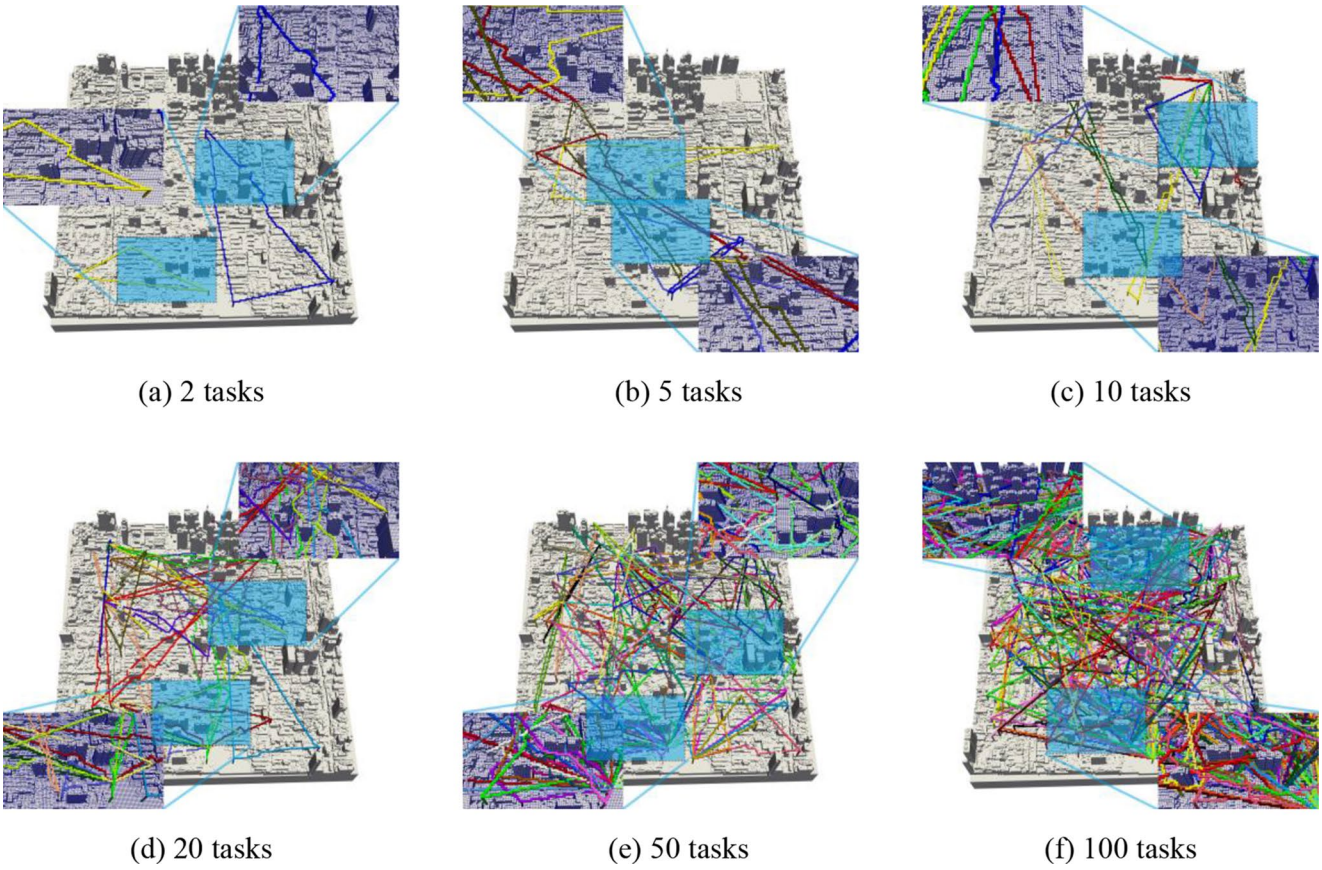
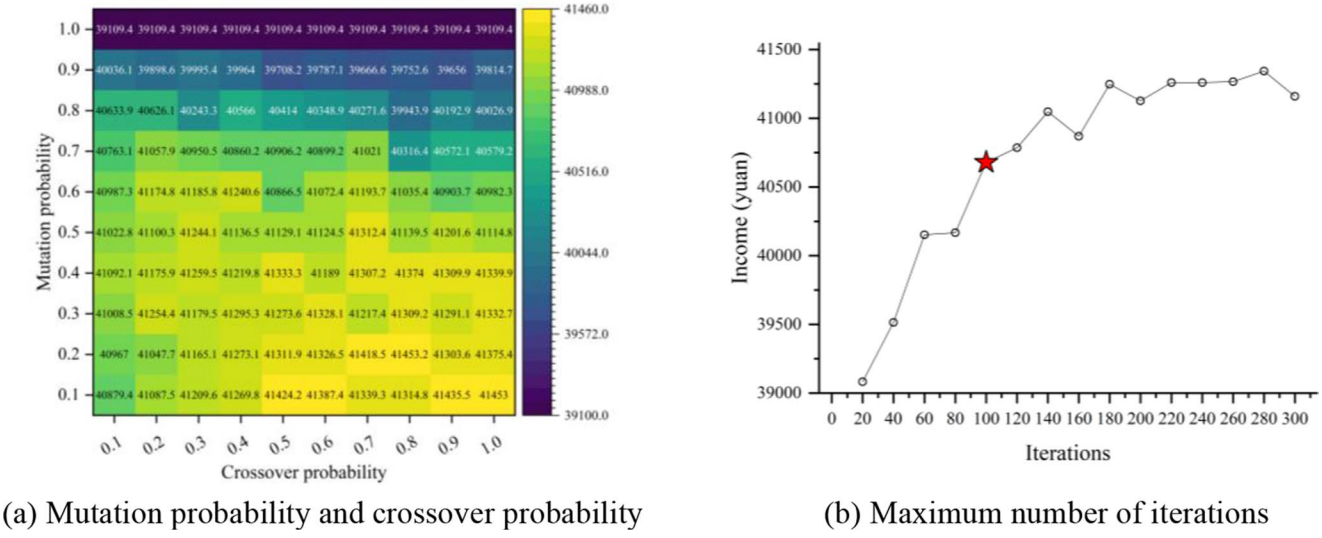



**Fig. 11** Route map of the UAV when performing an actual task




(a) Mutation probability and crossover probability (b) Maximum number of iterations

**Fig. 12** Parameter selection of TGALS

**Table 5** Total incomes (mean and variance) calculated for the robustness experiment (units: RMB)

| Number of tasks | Algorithm | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | HA | | OSDAA | | VsC_mH | | DMCHBA | | CNP | |
| | Mean | Var | Mean | Var | Mean | Var | Mean | Var | Mean | Var |
| 20 | 9577.89 | 0 | **9600.16** | **0** | 9600.00 | 0 | 9577.89 | 0 | 9577.89 | 0 |
| 40 | 17410.57 | 7.27E-12 | **17558.67** | **3.63E-12** | **17558.67** | **3.63E-12** | 17410.57 | 7.27E-12 | 17410.57 | 7.27E-12 |
| 60 | 24059.71 | 3.63E-12 | **24450.91** | **1.45E-11** | **24450.91** | **1.45E-11** | 24059.71 | 3.63E-12 | 24059.71 | 3.63E-12 |
| 80 | 29218.41 | 1.09E-11 | 29968.86 | 7.27E-12 | 29968.86 | 7.27E-12 | 29218.41 | 1.09E-11 | 29221.46 | 1.81E-11 |
| 100 | 35227.18 | 1.45E-11 | 36035.92 | 7.27E-12 | 36035.92 | 7.27E-12 | 35227.18 | 1.45E-11 | 35151.57 | 2.18E-11 |
| 120 | 32648.01 | 2.18E-11 | 36094.84 | 2.18E-11 | 36094.84 | 2.18E-11 | 9543.70 | 5.45E-12 | 35973.34 | 7.27E-12 |
| 140 | 33904.49 | 1.45E-11 | 37281.11 | 7.27E-12 | 37281.11 | 7.27E-12 | 17917.24 | 3.63E-12 | 37059.36 | 2.18E-11 |
| 160 | 35490.71 | 1.45E-11 | 39157.33 | 1.45E-11 | 39157.33 | 1.45E-11 | 25004.67 | 3.63E-12 | 39252.92 | 3.63E-12 |
| 180 | 36070.33 | 1.45E-11 | 40016.34 | 1.45E-11 | 40016.34 | 1.45E-11 | 31802.16 | 7.27E-12 | 39867.27 | 2.18E-11 |
| 200 | 34130.57 | 2.18E-11 | 39350.44 | 1.45E-11 | 39350.44 | 1.45E-11 | 34130.57 | 2.18E-11 | 39379.30 | 2.18E-11 |
| Rank-sum test | (10/0/0) | | (7/0/3) | | (8/0/2) | | (10/0/0) | | (10/0/0) | |
| 20 | 9563.57 | 9.93 | 9586.82 | 3.72 | 9587.50 | 3.74 | 9600.15 | 0.03 | **9600.16** | **0** |
| 40 | 17292.43 | 23.6 | 17397.17 | 17.31 | 17396.17 | 12.62 | 17549.94 | 3.95 | **17558.67** | **3.63E-12** |
| 60 | 23505.00 | 66.95 | 23847.87 | 45.44 | 23860.63 | 42.56 | 24354.36 | 17.97 | **24450.91** | **1.45E-11** |
| 80 | 28147.11 | 118.2 | 28677.10 | 68.8 | 28707.08 | 69.4 | 29575.08 | 58.8 | **29971.80** | **0.642** |
| 100 | 33007.50 | 122.8 | 33909.48 | 116.3 | 33913.24 | 139.9 | 35041.11 | 127.06 | **36081.82** | **1.023** |
| 120 | 31658.34 | 192.9 | 32726.93 | 123.29 | 32645.70 | 133.92 | 34044.89 | 246.1 | **36795.41** | **19.31** |
| 140 | 31982.98 | 144.7 | 33006.96 | 166.83 | 32982.86 | 153.66 | 34709.77 | 264.72 | **38686.52** | **32.68** |
| 160 | 32943.08 | 127.7 | 34072.08 | 170.06 | 34045.73 | 189.31 | 36107.60 | 325.51 | **40887.57** | **38.54** |
| 180 | 33421.16 | 173.78 | 34595.31 | 159.9 | 34580.45 | 206.65 | 36979.27 | 278.7 | **41678.38** | **40.9** |
| 200 | 32328.04 | 285.65 | 33446.43 | 153.97 | 33376.80 | 132.69 | 35980.03 | 367.83 | **41428.34** | **47.86** |
| Rank-sum test | (10/0/0) | | (10/0/0) | | (10/0/0) | | (10/0/0) | | | |

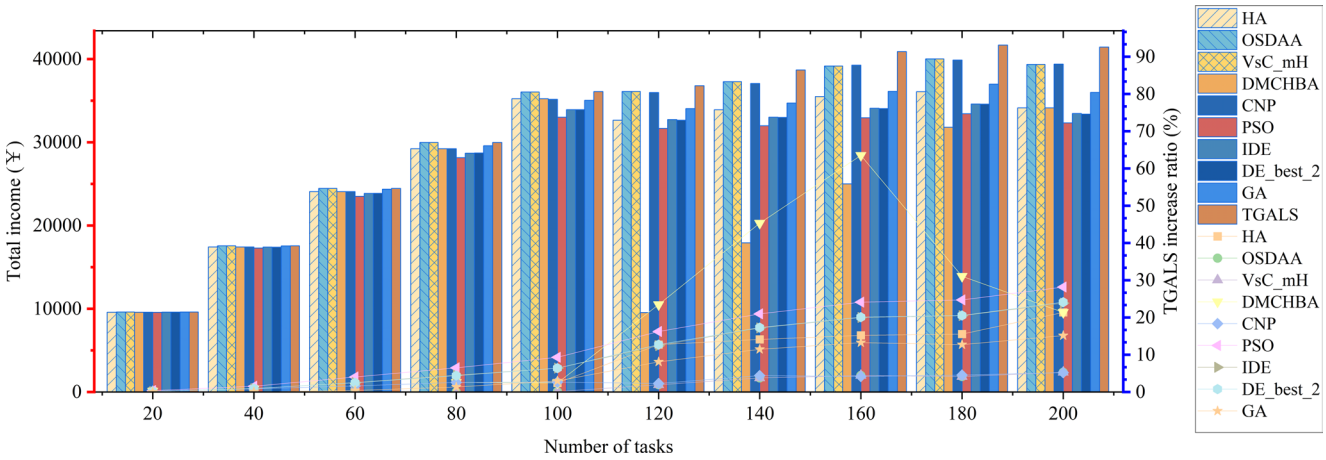



**Fig. 13** Total incomes (bars) calculated for the robustness experiment and the trends (lines) in the percentage increase in the income achieved by TGALS compared to the other algorithms

In general, the TGALS algorithm solved the task allocation scheme with the highest total revenue for different numbers of tasks, demonstrating its superior optimization performance.

### 4.4.4 High-efficiency experiment

Table 6 lists the mean and variance of the average computation times of the five iterative algorithms.

According to Table 6; Fig. 15, TGALS completed the allocation of 100 logistics tasks in 0.09 s, and that of 1000 tasks in 6.89 s, demonstrating its high calculation speed. It can be see from Fig. 15, the calculation time of TGALS was lower than that of GA. This proves that the calculation efficiency of TGALS significantly improved after using the local solution heuristic strategy, and further verifies the effectiveness of the two-stage optimization proposed in this paper.

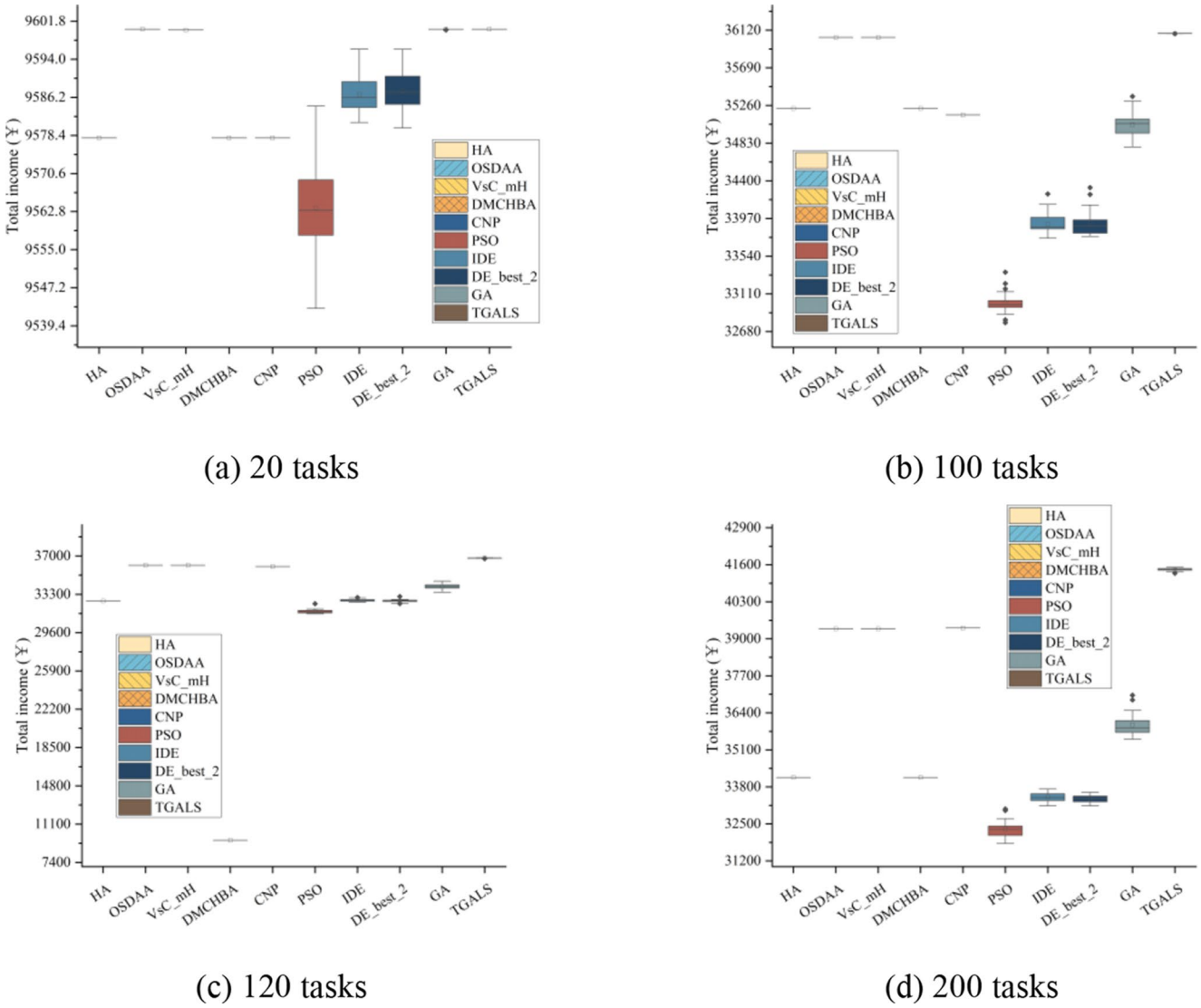



**Fig. 14** Box plots of the total revenues achieved by the algorithms in the robustness experiment

**Table 6** Computation times (mean and variance) calculated for the efficiency experiment (units: s)

| Number of tasks | Algorithm | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | PSO | | IDE | | DE_best_2 | | GA | | TGALS | |
| | Mean | Var | Mean | Var | Mean | Var | Mean | Var | Mean | Var |
| 100 | 0.1302 | 0.0035 | 0.1573 | 0.0042 | 0.1558 | 0.0027 | 0.1201 | 0.0040 | **0.0959** | **0.0039** |
| 200 | 0.3468 | 0.0063 | 0.3640 | 0.0068 | 0.3656 | 0.0072 | 0.2826 | 0.0068 | **0.1915** | **0.0042** |
| 300 | 0.7069 | 0.0092 | 0.7403 | 0.0088 | 0.7460 | 0.0075 | 0.6422 | 0.0102 | **0.4778** | **0.0076** |
| 400 | 1.2136 | 0.0189 | 1.2660 | 0.0129 | 1.2746 | 0.0147 | 1.1446 | 0.0139 | **0.9218** | **0.0101** |
| 500 | 1.8951 | 0.0169 | 1.9746 | 0.0214 | 1.9820 | 0.0184 | 1.8420 | 0.0150 | **1.5152** | **0.0149** |
| 600 | 2.7100 | 0.0359 | 2.7814 | 0.0314 | 2.8012 | 0.0297 | 2.6737 | 0.0449 | **2.2720** | **0.0366** |
| 700 | 3.7099 | 0.0235 | 3.7949 | 0.0311 | 3.8060 | 0.0288 | 3.7103 | 0.0272 | **3.2070** | **0.0306** |
| 800 | 4.7943 | 0.0706 | 4.9142 | 0.0440 | 4.9326 | 0.0511 | 4.8492 | 0.0687 | **4.2688** | **0.0322** |
| 900 | 6.0928 | 0.0455 | 6.2258 | 0.0412 | 6.2453 | 0.0461 | 6.1878 | 0.0603 | **5.5264** | **0.0446** |
| 1000 | 7.4718 | 0.0545 | 7.6543 | 0.0649 | 7.6726 | 0.0426 | 7.6680 | 0.1013 | **6.8955** | **0.0489** |

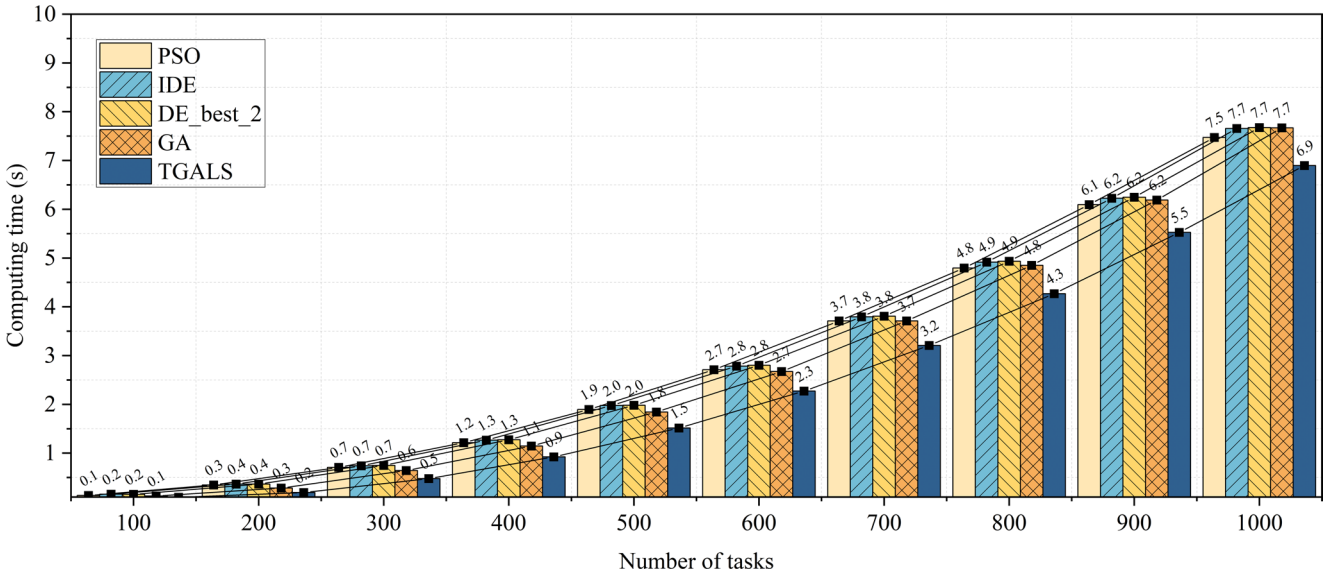



**Fig. 15** Calculate the time bar statistical diagram

**Table 7** Ablation experimental results statistics (mean, units: RMB)

| Number of tasks | Algorithm | | |
|---|---|---|---|
| | TGALS_A | TGALS_B | TGALS |
| 20 | 8628.689 | **8654.011** | **8654.011** |
| 40 | 16720.564 | 16991.473 | **17000.693** |
| 60 | 24008.043 | 24743.761 | **24845.21** |
| 80 | 28960.485 | 30532.922 | **30913.452** |
| 100 | 32310.329 | 34300.684 | **35403.355** |
| 120 | 32559.664 | 35099.242 | **38362.749** |
| 140 | 31874.500 | 34357.30 | **37947.519** |
| 160 | 32431.858 | 36051.120 | **40655.093** |
| 180 | 32442.822 | 35666.318 | **40799.121** |
| 200 | 31904.047 | 35864.501 | **41095.763** |
| Rank-sum test | (10/0/0) | (10/0/0) | |

In terms of the time complexity of the algorithm, according to the TGALS pseudo-code, the time complexity of the Hungarian algorithm is $O\left(n^3\right)$. Assuming that the population size is $ps$, the individual dimension is $n$, and the number of iterations is $T_{iter}$, the time complexity of the genetic algorithm is $O\left(T_{iter} \cdot ps \cdot n\right) = O(ps \cdot n)$. The population size $ps$ is usually proportional to the task size, so the time complexity of the improved genetic algorithm can be expressed as $O\left(n^2\right)$. Then the time complexity of TGALS is $O\left(n^3\right) = O\left(n^2\right) + O\left(n^3\right)$. In general, the analysis of the results of the feasibility, robustness, and efficiency experiments demonstrated that TGALS can optimize the multi-UAV task allocation scheme with a higher overall benefit in a shorter time frame, and can be effectively used as a solution algorithm for large-scale UAV task allocation problems.

#### 4.4.5 Ablation experiment

In this subsection, the ablation experiment of TGALS was carried out, where TGALS_A indicates that the two-stage optimization strategy was not adopted, and TGALS_B indicates that the conflict resolution strategy was not adopted. From the experimental results, it can be analyzed that the two strategies have a great effect on the optimization ability of TGALS. In large-scale task allocation, the two strategies play an important role together Table 7.

## 5 Conclusions

In this paper, to address the problem of multi-UAV task allocation, global performance optimization was performed by combining the heuristic Hungarian algorithm and the genetic algorithm with a conflict resolution mechanism. To achieve coupling optimization between the task allocation and path planning, the actual flight path cost of the UAVs was used as the evaluation standard for the task allocation. In addition, a dynamic correction strategy was designed to balance the global exploration and local exploitation abilities of the algorithm, enabling it to achieve an enhanced optimization performance. The proposed algorithm produced a superior solution and required a shorter calculation time. In general, the two-stage genetic algorithm proposed in this paper demonstrated a more advanced solution compared to mainstream algorithms and provided a fast, efficient, and stable optimization algorithm for allocating tasks to multiple low-altitude UAVs. In future work, we will expand the application scenarios of the algorithm and provide solutions for additional practical applications.

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**Author contributions** G.H. performed the algorithm innovation, designed the experiment, and wrote the paper. T.X. played an essential role in this work and was responsible for the direction of the research and content. S.Y., C.J., and Y.C. prepared and analyzed the experimental data. B.Y. proofread the manuscript. L.H. and G.C. were responsible for producing the relevant charts. The eight authors complemented each other in different ways and together supported the results presented in this paper.

**Data availability** No datasets were generated or analysed during the current study.

## Declarations

**Competing interests** The authors declare no competing interests.

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